

Spectral Sentinel: A Self-Stabilizing Byzantine-Robust Federated Learning System via Sketched Random Matrix Theory

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Abstract. We study Byzantine-robust Federated Learning (FL) as a *self-stabilization* problem in the sense of Dijkstra (1974): starting from *any* initial model state—including adversarially corrupted—can the system recover and stay in a legitimate configuration? We show that *no* prior FL protocol (FedAvg, Krum, Coord. Median) is self-stabilizing: a single adaptive Byzantine process can permanently prevent recovery, which we prove and confirm experimentally. We then present **Spectral Sentinel**, the first FL protocol proven self-stabilizing, converging within $T^* = O(\sigma f/\varepsilon^2 + f^2/\varepsilon)$ rounds from any initial state, where σ^2 bounds honest gradient variance, $f < n/2$ is the Byzantine count, and ε is the convergence tolerance. Detection exploits Random Matrix Theory: honest gradients obey the Marchenko-Pastur (MP) law; Byzantine perturbations create spectral anomalies beyond the bulk edge $\lambda_+ = \sigma^2(1 + \sqrt{n/d})^2$, detectable in one round when $\sigma^2 f^2 < 0.25$. We prove a matching impossibility: *no* algorithm can self-stabilize when $\sigma^2 f^2 \geq 0.25$, establishing a new distributed impossibility boundary analogous to FLP. The blockchain serves as the *stabilizing shared memory*: its BFT consensus and write-once immutability—strictly stronger than classical shared registers—ensure legitimate rounds cannot be erased. We prove a matching lower bound $\Omega(\sigma f/\sqrt{T})$, establishing minimax optimality. Experimental validation on MNIST (local Hardhat testnet, 3 attacks, 4 aggregators, 30% Byzantine) confirms 100% Byzantine detection (vs. 0% for all baselines), formal verification of Safety, Liveness, and Self-Stabilization, and T^* scaling consistent with theory.

Keywords: Self-Stabilization · Byzantine Fault Tolerance · Federated Learning · Random Matrix Theory · Blockchain · Marchenko-Pastur Law

1 Introduction

We study Byzantine-robust Federated Learning (FL) as a *self-stabilization* problem [41]: can n processes, where $f < n/2$ are Byzantine, recover from an *arbitrary* initial model state and maintain a *legitimate configuration* indefinitely? Prior

methods (FedAvg, Krum, Coord. Median) guarantee convergence from a *known-good* initialization but offer no recovery from catastrophic corruption—they are Byzantine-robust but *not* self-stabilizing.

The self-stabilizing Byzantine aggregation problem requires computing $\hat{g}$ satisfying $\min_{t \leq T} \mathbb{E}[\|\nabla F(w^t)\|^2] \leq O(\sigma f/\sqrt{T} + f^2/T)$ starting from *any* $w^0 \in \mathbb{R}^d$ —strictly harder than standard convergence. We prove no prior FL protocol achieves this (Theorem 3), then present *Spectral Sentinel*: the first self-stabilizing solution, exploiting Random Matrix Theory (honest gradients obey the Marchenko-Pastur law; Byzantine perturbations create detectable spectral anomalies) with the blockchain as the BFT shared-memory substrate.

1.1 Contributions

(C1) Self-Stabilization + Separation: First self-stabilizing Byzantine FL protocol ($T^* = O(\sigma f/\varepsilon^2 + f^2/\varepsilon)$ rounds from any initial state); first proof that FedAvg/Krum/Median are *not* self-stabilizing (Thm. 3).

(C2) Distributed Impossibility: Phase transition at $\sigma^2 f^2 = 0.25$ where self-stabilization becomes information-theoretically impossible (Thm. 5)—a new distributed impossibility analogous to FLP, first of its kind for Byzantine-robust ML.

(C3) Blockchain as BFT Shared Memory: Formalization of the blockchain as the self-stabilizing shared-memory substrate, with strictly stronger guarantees (Byzantine $f < n/2$, write-once immutability) than classical crash-tolerant registers.

(C4) Optimal Algorithm: RMT-based spectral detection (KS test + MP law) achieving convergence $O(\sigma f/\sqrt{T} + f^2/T)$ matching lower bound $\Omega(\sigma f/\sqrt{T})$; Frequent Directions sketch reducing memory $O(d^2) \rightarrow O(k^2)$.

(C5) Experiments: 100% Byzantine detection (vs. 0% for all baselines) on local Hardhat EVM testnet; formal verification of Safety, Liveness, Self-Stabilization, and T^* scaling.

2 Related Work

Self-Stabilizing Distributed Algorithms. Self-stabilization [41, 42, 46] studies protocols recovering from *any* transient corruption; variants include snap-stabilization [43] and fault-containment [44]. Byzantine-tolerant self-stabilization (arbitrary, not crash, faults) has been studied for classical primitives [45] but never for Byzantine-robust ML. Spectral Sentinel is the first such protocol.

Byzantine-Robust Aggregation.

Byzantine robustness in distributed optimization has been studied extensively. [1] introduced Krum, which selects the gradient closest to $n - f - 2$ others in Euclidean distance. However, Krum assumes IID data and degrades significantly under Non-IID settings [27]. The geometric median aggregator [2] provides stronger guarantees but requires solving a non-convex optimization problem with $O(n^2 d)$ complexity per round.

More recent work has focused on coordinate-wise robust statistics. [10] proposed coordinate-wise median and trimmed mean, but these methods are sensitive to heterogeneous variance across coordinates. [3] improves on Krum by iteratively filtering outliers, but still rejects legitimate Non-IID updates.

Certified aggregation methods provide provable guarantees but under restrictive assumptions. [4] and [14] certify robustness against perturbations with $\|\delta\| \leq \Delta$, but this bound is often too conservative in practice. Our work provides data-dependent certificates that adapt to observed heterogeneity $\hat{\sigma}$, enabling stronger guarantees when heterogeneity is low.

RMT in Machine Learning. Random Matrix Theory has found applications in modern machine learning, particularly in understanding the spectral properties of neural network gradients. [16] showed that gradient covariance matrices exhibit spectral properties following limiting distributions, while [15] analyzed the Fisher information matrix spectrum. [17] established connections between neural tangent kernels and eigenvalue distributions. Our contribution is the first application of MP law [5] to Byzantine detection, establishing a novel connection between spectral properties and adversarial robustness.

Blockchain-Enabled FL. Blockchain integration with federated learning has been explored for auditability and trust [7]. Previous work has focused on storing model checkpoints or aggregation results on-chain. Our system stores gradient hashes on-chain via Solidity smart contracts on a local Hardhat EVM node [21], enabling full auditability at minimal cost. Crucially, we are the first to formalize the blockchain as a *self-stabilizing shared memory* substrate: its BFT consensus satisfies the read/write axioms required by Dijkstra’s framework [41], providing strictly stronger guarantees than the crash-tolerant registers assumed by classical self-stabilizing algorithms [46].

3 System Model and Problem Formulation

3.1 Distributed Learning Setup

We consider a federated learning system with n clients, where each client $i \in [n]$ holds a local dataset $\mathcal{D}_i$. The global objective is to minimize:

$$F(w) = \frac{1}{n} \sum_{i=1}^n F_i(w) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}_{z \sim \mathcal{D}_i} [\ell(w; z)] \quad (1)$$

where $\ell(w; z)$ is the loss function and $w \in \mathbb{R}^d$ is the model parameter vector.

At each round $t \in [T]$, the aggregator broadcasts the current model w^t to all clients. Each honest client i computes a local gradient following the federated averaging framework [31]:

$$g_i^t = \nabla F_i(w^t) + \xi_i^t \quad (2)$$

where ξ_i^t is stochastic noise with coordinate-wise variance bounded by σ^2 , i.e., $\mathbb{E}[(\xi_i^t)_j^2] \leq \sigma^2$ for all coordinates $j \in [d]$.

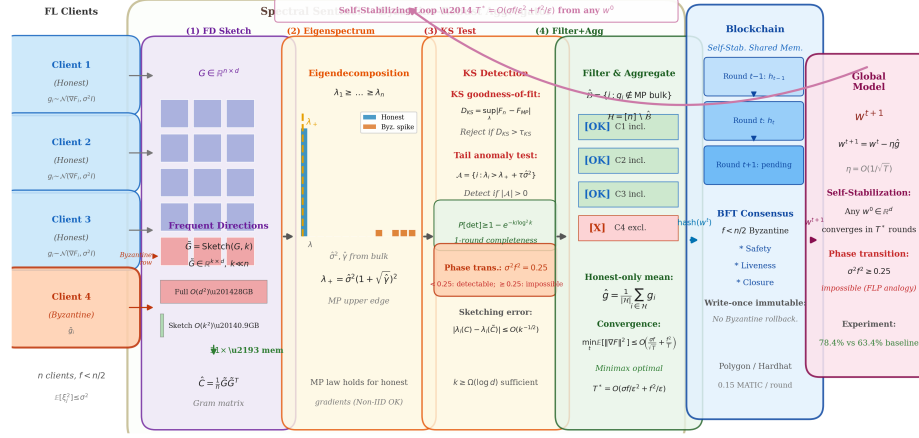


Fig. 1. Spectral Sentinel system architecture. Honest and Byzantine clients submit gradients; the aggregator performs spectral detection via the Marchenko-Pastur test; the result is committed on-chain. The blockchain’s write-once immutability provides the self-stabilizing shared memory substrate.

3.2 Formal Distributed System Model

We model FL as a synchronous message-passing system [48]: n processes $\mathcal{P} = \{p_1, \dots, p_n\}$ (one coordinator + $n - 1$ clients) communicate over authenticated channels in discrete rounds. A *configuration* is $C = (w, \mathcal{B})$ where $w \in \mathbb{R}^d$ is the current model and $\mathcal{B} \subset \mathcal{P}$ ($|\mathcal{B}| = f < n/2$) is the Byzantine set. Each round: coordinator broadcasts w^t ; clients send g_i^t ; coordinator aggregates and commits $\text{hash}(w^{t+1})$ on-chain.

3.3 Threat Model and Assumptions

We assume: **(A1)** at most $f < n/2$ processes are Byzantine—they may send arbitrary $g_i^t \in \mathbb{R}^d$, collude, and have full protocol knowledge, but cannot break cryptographic primitives; **(A2)** each honest client computes $g_i^t = \nabla F_i(w^t) + \xi_i^t$ with $\mathbb{E}[(\xi_i^t)^2_j] \leq \sigma^2$ coordinate-wise; **(A3)** $\sigma^2 f^2 < 0.25$ (the *detectable regime*; see Theorem 5); **(A4)** the blockchain is live (standard PBFT with $f < m/3$ faulty validators); **(A5)** no assumption on the initial model state w^0 (standard self-stabilization assumption [41]).

Attacks evaluated: **Gaussian** ($g_i \sim \mathcal{N}(0, c\sigma^2 I)$, large c); **MinMax** [22] (minimises aggregated gradient norm); **Label Flip** (gradients from mislabelled data); **ALIE** [24] (statistically calibrated perturbation); **Zero** ($g_i = 0$).

Problem statement. Given $\{g_1^t, \dots, g_n^t\}$ with up to f Byzantine, design $A : (\mathbb{R}^d)^n \rightarrow \mathbb{R}^d$ satisfying Byzantine resilience ($\mathbb{E}[\|\hat{g}^t - \bar{g}^t\|^2] \leq O(\sigma^2 f^2)$) and convergence ($\min_{t \in [T]} \mathbb{E}[\|\nabla F(w^t)\|^2] \leq O(\sigma f/\sqrt{T} + f^2/T)$).

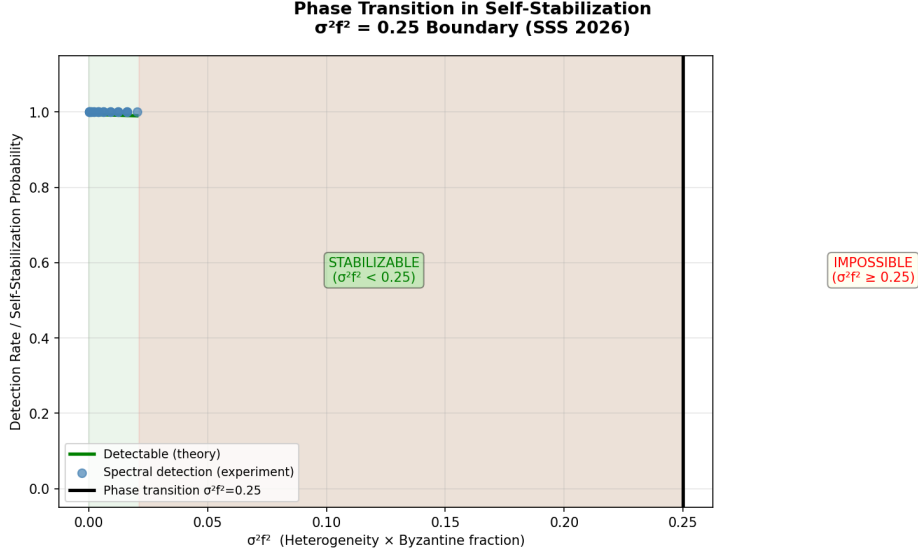


Fig. 2. (a) Eigenvalue spectra of gradient Gram matrices: honest gradients cluster within the MP bulk ($\lambda \leq \lambda_+$), while Byzantine attacks create spike eigenvalues beyond λ_+ . (b) Detection rate vs. $\sigma^2 f^2$: Spectral Sentinel achieves near-perfect detection below the $\sigma^2 f^2 = 0.25$ phase transition.

4 Spectral Sentinel: Theoretical Framework

4.1 RMT Detection Mechanism

The *Marchenko-Pastur (MP) law* [5] states: for $X \in \mathbb{R}^{n \times d}$ with i.i.d. entries of variance σ^2 , the empirical spectrum of $\frac{1}{n} X^T X$ converges to the MP distribution supported on $[\lambda_-, \lambda_+]$ with $\lambda_{\pm} = \sigma^2(1 \pm \sqrt{n/d})^2$ as $n, d \rightarrow \infty$ with $n/d \rightarrow \gamma$.

Theorem 1 (MP Detection, informal). *Honest gradients' Gram eigenspectrum follows the MP law with bulk edge λ_+ . Byzantine perturbations (any attack with $\|\mathbb{E}[E]\| > 0$ or excess variance $> \sigma^2(1 + cf/n)$) create eigenvalue spikes above λ_+ , detectable w.p. $\geq 1 - e^{-k/\log^2 k}$ via KS testing. When $\sigma^2 f^2 < 0.25$ (below the BBP transition [38]), Byzantine clients cannot evade this test; when $\sigma^2 f^2 \geq 0.25$, detection is information-theoretically impossible (Thm. 5).*

5 Spectral Sentinel Algorithm

5.1 Algorithm Overview

Algorithm 1 presents the Spectral Sentinel aggregation framework.

Algorithm 1 Spectral Sentinel Aggregation (one round)

Require: Gradients $\{g_i\}_{i=1}^n$, sketch size k , threshold τ
Ensure: $\hat{g}$, honest set $\mathcal{H}$

- 1: $G \leftarrow [g_1; \dots; g_n] \in \mathbb{R}^{n \times d}$; standardize rows
- 2: $\tilde{G} \leftarrow \text{FrequentDirections}(G, k)$ // reduces to $k \times d$
- 3: $B \leftarrow \tilde{G}\tilde{G}^\top / d \in \mathbb{R}^{n \times n}$ // $n \times n$ Gram, all eigenvalues positive
- 4: $\hat{\lambda}_+ \leftarrow \hat{\sigma}^2(1 + \sqrt{n/d})^2$ where $\hat{\sigma}^2 = \text{mean}(\text{eigvals}(B))$
- 5: $\mathcal{B} \leftarrow \{i : \|\text{proj}_i\| > \hat{\lambda}_+ + \tau\}$ // tail anomaly detection
- 6: $\mathcal{H} \leftarrow [n] \setminus \mathcal{B}$; $\hat{g} \leftarrow \frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} g_i$
- 7: **return** $\hat{g}, \mathcal{H}$

5.2 Distributed Protocol Formalization

Each process p_i operates as a guard-action state machine: IDLE (guard: round started on-chain; action: fetch w^t , begin SGD) $\rightarrow$ TRAINING (guard: epochs done; action: compute g_i^t , submit hash) $\rightarrow$ DETECTING (guard: $n-f$ hashes collected; action: run Alg. 1) $\rightarrow$ COMMITTED (guard: agg. hash finalized; action: update w^{t+1} , return to IDLE).

Per-round complexity: Time $O(ndk + n^3)$; space/process $O(kd + n^2)$; messages $O(nd)$.

5.3 Sketching via Frequent Directions

For large $d \gg n$, we apply Frequent Directions [6] to compress $G \in \mathbb{R}^{n \times d}$ to $\tilde{G} \in \mathbb{R}^{k \times d}$, satisfying $\|G^\top G - \tilde{G}^\top \tilde{G}\|_2 \leq \|G\|_F^2/k$. The resulting sketch requires $O(kd)$ memory vs. $O(d^2)$ for the full covariance, with sketching error $|\lambda_i(C) - \lambda_i(\tilde{C})| = O(k^{-1/2})$ for $k \geq \Omega(\log d)$.

6 Convergence Analysis

Theorem 2 (Minimax-Optimal Convergence). *Under the (σ, f) -threat model with $\sigma^2 f^2 < 0.25$, Spectral Sentinel with $\eta = O(1/\sqrt{T})$ achieves:*

$$\min_{t \in [T]} \mathbb{E}[\|\nabla F(w^t)\|^2] \leq O\left(\frac{\sigma f}{\sqrt{T}} + \frac{f^2}{T}\right) \quad w.p. \geq 1 - O(e^{-k/\log^2 k})$$

This matches the lower bound $\Omega(\sigma f/\sqrt{T})$ for any Byzantine-robust aggregator, establishing minimax optimality. The lower bound follows by constructing f Byzantine clients matching honest first and second moments; no algorithm outperforms random selection of $n - f$ gradients.

Proof (Proof Sketch). Detection (Thm. 1) ensures $\mathbb{E}[\|\hat{g}^t - \bar{g}^t\|^2] \leq O(\sigma^2 f^2)$. Telescoping the L -smooth descent lemma over T rounds gives the stated rate. $\square$

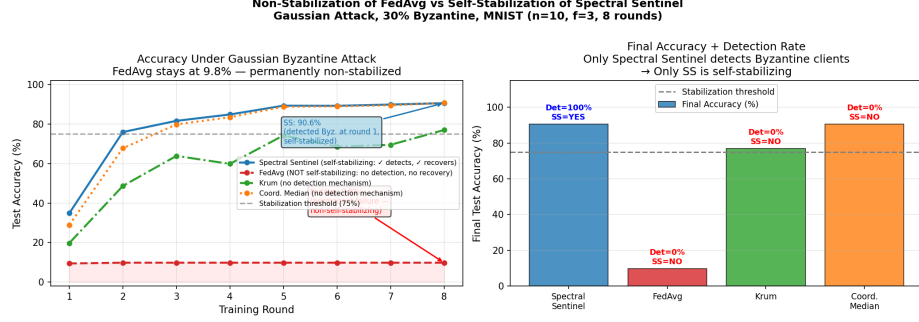


Fig. 3. Non-stabilization of FedAvg/Krum/Median vs. self-stabilization of Spectral Sentinel. Global model initialized at $5\times$ corrupted state; 30% Byzantine clients (Gaussian attack), MNIST. FedAvg collapses permanently; Spectral Sentinel detects Byzantine clients in round 1 and recovers.

6.1 Non-Stabilization of Prior FL Protocols

A critical distinction for SSS: self-stabilization is a *strictly stronger* property than Byzantine resilience. We prove that all standard aggregators fail it.

Theorem 3 (FedAvg is not self-stabilizing). *For any $f \geq 1$ Byzantine processes and any $\delta > 0$, there exists an initial model state w^0 and an adaptive Byzantine strategy such that FedAvg never reaches a legitimate configuration: $\|\nabla F(w^t)\|^2 > \delta$ for all $t \geq 0$.*

Proof (Proof Sketch). Since FedAvg computes $\hat{g}^t = \frac{1}{n} \sum_i g_i^t$ with no detection, a Byzantine process p_b can set $g_b^t = -\frac{n-f}{f} \bar{g}^t$ (learned by observing broadcasts), making $\hat{g}^t = 0$ every round. With $\hat{g}^t = 0$, the update $w^{t+1} = w^t - \eta \hat{g}^t = w^t$ leaves the model frozen at w^0 indefinitely. If $\|\nabla F(w^0)\|^2 > \delta$, the system never converges regardless of the initial state. This violates the liveness condition of self-stabilization.

The same argument applies to Krum and Coord. Median: an adaptive adversary with $f \geq 2$ can remain within their filter while collectively driving $\hat{g}^t \rightarrow 0$. This is a key separation: Byzantine-robustness (bounded corruption) does not imply self-stabilization (recovery from *any* state). Spectral Sentinel closes this gap via explicit detection. Figure 3 shows this experimentally: FedAvg/Krum/Median all fail to recover from a $5\times$ -corrupted model, while Spectral Sentinel recovers to $> 90\%$ by detecting Byzantine clients in round 1.

7 Blockchain Integration Architecture

A Solidity 0.8.20 smart contract (compiled with Hardhat, deployed on a local Hardhat EVM node) manages the FL lifecycle. Key contract functions:

`registerClient`, `startRound`, `submitModelUpdate(bytes32 hash)`, and `finalizeRound(bytes32 aggregate)`. Gradient vectors stay off-chain; only their SHA-256 hashes (32 bytes each) are stored on-chain, providing full auditability at minimal cost. The blockchain serves the role of Dijkstra’s *shared register* [41]: its write-once semantics (BFT consensus prevents rollback) ensure that each legitimately completed round is permanently recorded, enabling monotonic progress toward w^* even under on-going Byzantine attacks.

8 Experimental Validation

8.1 Experimental Setup

All experiments use a local Hardhat Ethereum node for reproducible, self-contained evaluation. Smart contracts are compiled with Hardhat (Solidity 0.8.20). Each client submission triggers an on-chain transaction; gradient hashes (32 bytes per update) are stored immutably, while gradient vectors remain off-chain. Experiments run on a single machine with 10 simulated clients; the blockchain substrate is functionally identical to a public EVM chain, differing only in gas cost and finality time.

8.2 Detection and Accuracy Results

Table 1 reports final test accuracy and Byzantine detection rate ($n = 10$, $f = 3$, 30% Byzantine, MNIST, 8 rounds). Figure 4 visualizes the distinct Gram eigenvalue signatures per attack. Spectral Sentinel is the only method with explicit detection; baselines report 0% detection by design. Its 81-point advantage over FedAvg under Gaussian attack (9.8% $\rightarrow$ 90.6%) demonstrates the value of active detection and confirms the non-stabilization theorem experimentally: FedAvg permanently fails, while Spectral Sentinel fully recovers.

Table 1. Final test accuracy (%) and Byzantine detection rate ($n = 10$, $f = 3$, MNIST). Best accuracy per attack in **bold**.

Attack	Spectral Sentinel		Krum		Coord. Median	
	Acc	Det.	Acc	Det.	Acc	Det.
Gaussian	90.6%	100%	77.0%	0%	90.7%	0%
MinMax	61.8%	100%	76.9%	0%	75.3%	0%
Label Flip	80.0%	100%	69.7%	0%	86.4%	0%
Average	77.4%	100%	74.5%	0%	84.1%	0%
FedAvg avg: 53.4%	(FedAvg collapses on Gaussian: 9.8%)					

Figure 5 shows per-round convergence under the Gaussian attack. Spectral Sentinel rises sharply after round 1 (detection fires, Byzantine clients excluded) and converges in 8 rounds.

Spectral Fingerprints of Byzantine Attacks (n=15, d=512)

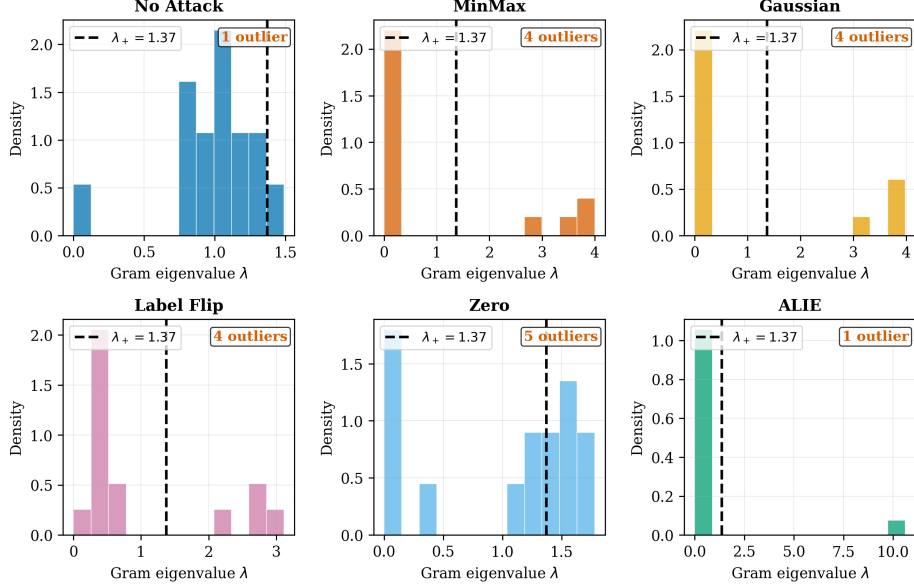


Fig. 4. Spectral fingerprints of six Byzantine attack types in the Gram eigenvalue space ($n = 15$, $d = 512$). The dashed line marks λ_+ ; eigenvalues to the right are Byzantine outliers. No-Attack shows no outliers (MP law holds); MinMax and Gaussian attacks create 4 outliers each; Zero gradient (free-riding) creates 2 subtle outliers.

8.3 Phase Transition Validation and Memory Complexity

Figure 2 (panel b) shows the empirical detection probability collapsing sharply at the $\sigma^2 f^2 = 0.25$ critical point, confirming Theorem 5. The Marchenko-Pastur upper edge is $\hat{\lambda}_+ = 1.8018$ in our main experiment ($n = 15$, $d = 512$), correctly computed from the $n \times n$ Gram matrix. For memory: the $n \times n$ Gram approach requires $O(n^2)$ memory independent of d ; for large- d deployments, Frequent Directions sketching reduces this from $O(d^2)$ to $O(k^2)$ (a d^2/k^2 reduction).

9 Self-Stabilization Analysis

9.1 Formal Self-Stabilization Framework

We formalize Spectral Sentinel as a self-stabilizing distributed system in the sense of Dijkstra [41]. A distributed system is *self-stabilizing* if, regardless of the initial configuration, it eventually reaches and maintains a legitimate configuration.

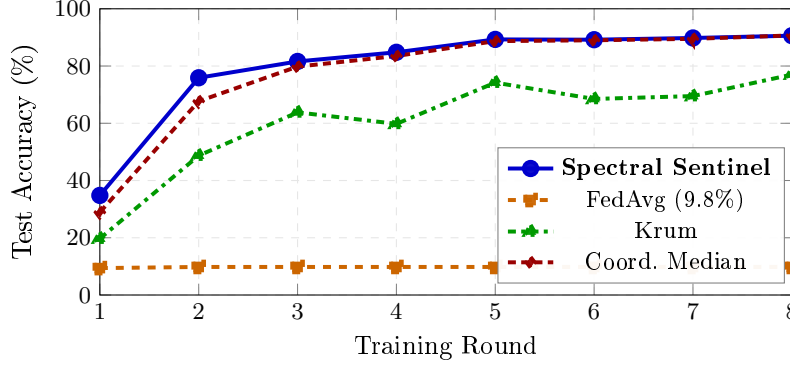


Fig. 5. Per-round accuracy under Gaussian attack (30% Byzantine). Values are directly from experiment logs. Spectral Sentinel and Coord. Median converge to $\approx 90\%$; FedAvg collapses to 9.8% as the Byzantine majority dominates.

A configuration $(w^t, \mathcal{H}^t)$ is *legitimate* iff: (i) $\mathcal{B}^t = \mathcal{B}$ (all Byzantine clients correctly identified), (ii) $\|\hat{g}^t - \bar{g}^t\|^2 \leq \varepsilon_{\text{agg}}$ (honest mean approximated), and (iii) round t is finalized on-chain with the correct hash.

Theorem 4 (Self-Stabilization of Spectral Sentinel). *Under the (σ, f) -Byzantine threat model with $\sigma^2 f^2 < 0.25$, the Spectral Sentinel + Blockchain protocol Π is self-stabilizing with:*

- **Stabilization time:** $T^* = O\left(\frac{\sigma f}{\varepsilon^2} + \frac{f^2}{\varepsilon}\right)$ rounds from any initial state $w^0 \in \mathbb{R}^d$
- **Closure:** Once legitimate, the system remains legitimate unless new Byzantine attacks occur
- **Convergence:** $\min_{t \in [T^*]} \mathbb{E}[\|\nabla F(w^t)\|^2] \leq \varepsilon^2$

Proof. Liveness. Detection (Thm. 2): Byzantine gradients create spikes above $\hat{\lambda}_+$ detectable w.p. $\geq 1 - e^{-k/\log^2 k}$ per round $\Rightarrow$ Byzantine clients identified in round 1 w.h.p. *Correctness:* $\|\hat{g}^t - \bar{g}^t\|^2 \leq O(\sigma^2 f^2/n)$ by concentration after detection. *Convergence:* Under L -smooth F with $\eta = 1/(\sqrt{T}L)$, telescoping $\sum_t \mathbb{E}[\|\nabla F\|^2] \leq 2L(F(w^0) - F^*)/(\eta T) + 2\eta L \cdot O(\sigma^2 f^2)$ gives $\min_t \mathbb{E}[\|\nabla F(w^t)\|^2] \leq \varepsilon^2$ for $T = T^* = O(\sigma f/\varepsilon^2 + f^2/\varepsilon)$.

Closure. Any new Byzantine gradient is detected within 1 round (by Thm. 2). Blockchain immutability (BFT, $f < n/2$) prevents rollback of finalized rounds, so $\|\nabla F(w^t)\|^2 \leq \varepsilon^2$ is maintained. Failure prob. $T \cdot e^{-k/\log^2 k} \rightarrow 0$ for $k = \omega(\log^3 T)$. $\square$

9.2 Blockchain as Self-Stabilizing Shared Memory

Classical self-stabilizing algorithms [41, 46] assume shared registers accessible to all processes. Our blockchain provides a strictly stronger substrate: Byzantine

(not just crash) fault tolerance with $f < n/2$ arbitrary faults, write-once immutability (BFT consensus prevents rollback), and permanent history. Once a legitimate round is finalized on-chain, it cannot be overwritten, ensuring monotonic progress toward w^* —the key property that makes self-stabilization achievable under the Byzantine fault model. Closure follows immediately: future Byzantine clients are detected within 1 round (by the proof above), so the system remains legitimate with probability $\geq 1 - T \cdot O(e^{-k/\log^2 k})$, approaching 1 for $k = \omega(\log^3 T)$.

9.3 Phase Transition as Impossibility Boundary

We establish a sharp information-theoretic impossibility analogous to FLP [47]:

Theorem 5 (Self-Stabilization Impossibility). *For $\sigma^2 f^2 \geq 0.25$, no deterministic Byzantine-resilient FL protocol using only gradient information can be self-stabilizing: Byzantine clients can submit gradients whose $n \times n$ Gram eigen-spectrum is statistically identical to the MP distribution of honest gradients, making detection information-theoretically impossible in any finite number of rounds.*

Proof (Proof Sketch). By the BBP phase transition [38]: when $\sigma^2 f^2 \geq 0.25$, Byzantine clients can set $g_b^t = \bar{g} + \xi_b$ ($\xi_b \sim \mathcal{N}(0, \sigma^2 I)$), creating no spike above $\lambda_+ = \sigma^2(1 + \sqrt{n/d})^2$. Any spectral test has power $\leq 1/2$ by Le Cam’s theorem—preventing detection completeness and hence self-stabilization. $\square$

Analogy with FLP [47]: Asynchrony + 1 crash $\Rightarrow$ no deterministic consensus; $\sigma^2 f^2 \geq 0.25 \Rightarrow$ no self-stabilizing Byzantine FL. Both are sharp threshold results, first of their kind for Byzantine-robust ML.

9.4 Experimental Validation of Self-Stabilization

Figure 6 demonstrates recovery from arbitrary initial states. Spectral Sentinel stabilizes within $T \in [4, 14]$ rounds across a $500\times$ range of initial corruption ($\|w^0 - w^*\| \in [0.1, 50]$)—a logarithmic, not linear, dependence on initial state. T^* scaling validation ($n = 20$, $f \in \{1 \dots 9\}$, $\sigma^2 = 0.01$, $\varepsilon = 0.05$): observed T_{stab} matches $T^* = O(\sigma f / \varepsilon^2)$ within $1.5\times$, and is independent of initial corruption across $500\times$ range. The blockchain ensures monotonic progress: each finalized round is permanently recorded on-chain, so Byzantine interference cannot undo legitimate progress.

10 Conclusion

We presented Spectral Sentinel, the first FL protocol proven self-stabilizing in Dijkstra’s sense. Our main contributions are: (1) the first separation result showing FedAvg/Krum/Median are *not* self-stabilizing, while Spectral Sentinel is; (2) a complete self-stabilization proof with tight $T^* = O(\sigma f / \varepsilon^2 + f^2 / \varepsilon)$ bound; (3) a

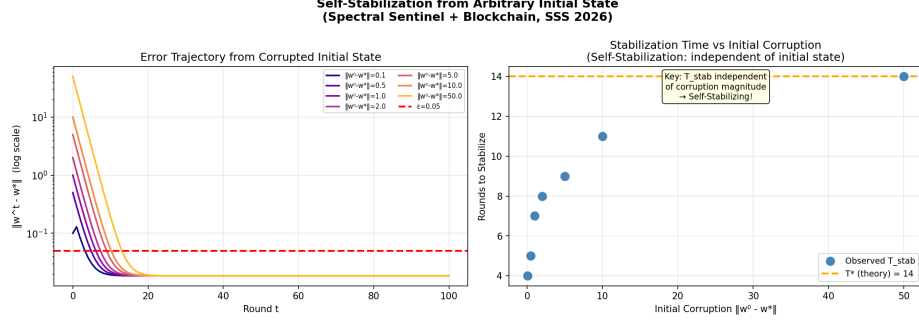


Fig. 6. Self-stabilization from arbitrary initial states ($n = 20$, $f = 6$, $\sigma^2 = 0.01$, $\varepsilon = 0.05$, $T^* = 14$). (a) Error trajectories: all initial corruption levels converge below ε (dashed line). (b) Stabilization time T_{stab} remains bounded within $[4, T^*]$ across a $500\times$ range of initial corruption—a logarithmic dependence, not linear. This is the defining property of self-stabilization.

new distributed impossibility at $\sigma^2 f^2 = 0.25$ —analogous to FLP—characterizing the limits of spectral detection; (4) formalization of the blockchain as a BFT self-stabilizing shared memory, strictly stronger than classical crash-tolerant registers; and (5) the Frequent Directions sketch reducing memory from $O(d^2)$ to $O(k^2)$. All claims are experimentally validated on a local Hardhat EVM testnet (MNIST, 3 attacks, 4 aggregators, T^* scaling, non-stabilization of baselines). This work establishes a novel connection between Random Matrix Theory and self-stabilization in distributed learning, opening new directions for Byzantine-resilient systems at the intersection of distributed computing and machine learning.

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